CardioLens 3D

Coronary artery disease and per-vessel stenosis risk, mapped onto a real 3D heart, with the reasons behind every number.

Multimodal AI Hackathon 2026 · IIT Mandi · Track A: Cardiovascular Risk Visualization & Prediction · Ashwin Pareek

Clinical safety. CardioLens is for decision support and education only. Its predictions are not a diagnosis and do not replace coronary angiography, CT coronary angiography or a clinician's judgement. The same notice is permanently visible in the application header and on every exported report.

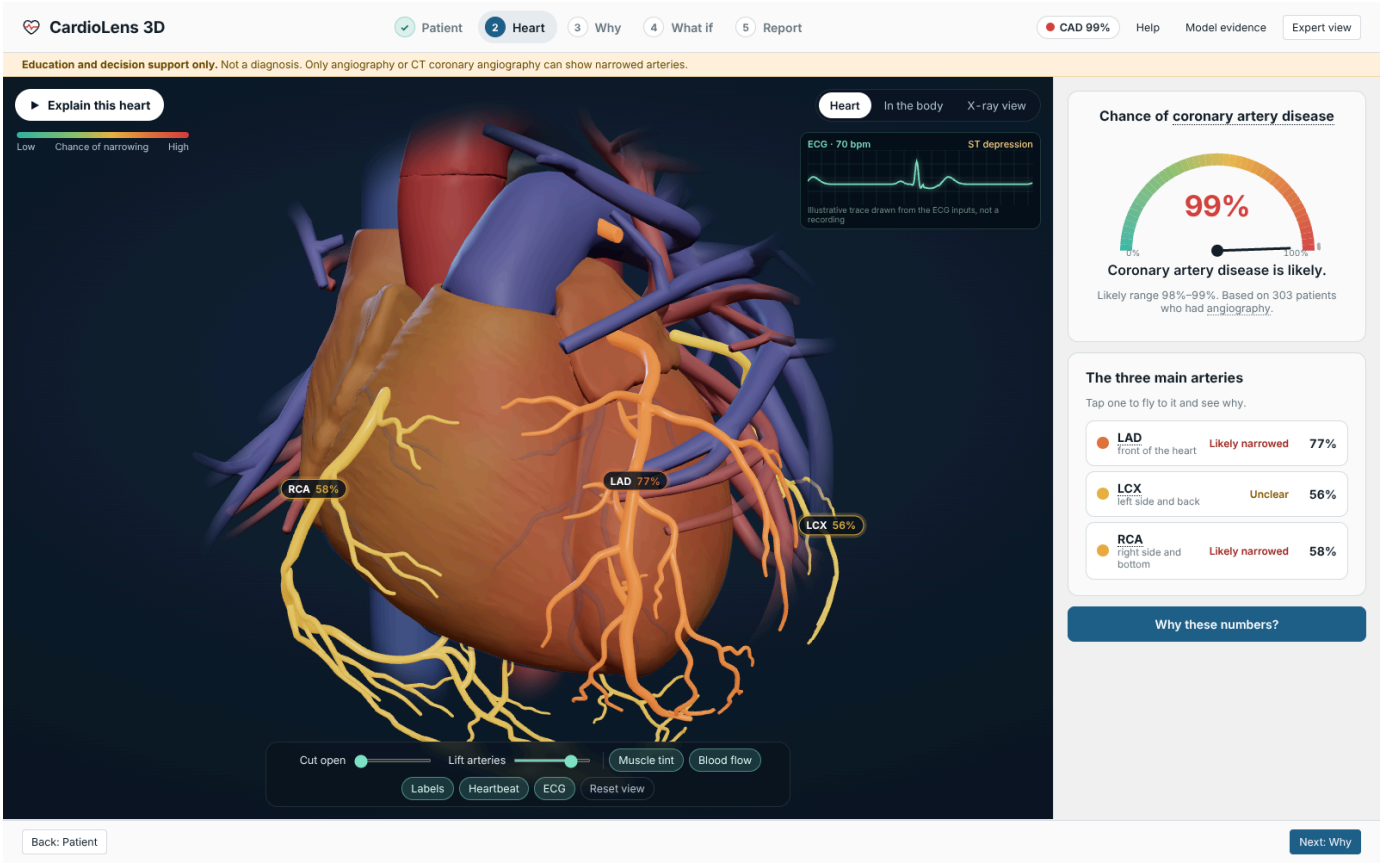


Figure 1. Guided view, step 2 of 5: the BodyParts3D heart with arteries lifted off the surface (exploded view), each coloured by its own predicted stenosis probability, a live ECG drawn from the ECG inputs, the CAD gauge with its 90% range, and plain-language artery verdicts. An Expert view shows every panel at once.

1. Problem and approach

Risk calculators return a single percentage, but cardiologists reason in arteries and territories. Track A asks for a system that predicts overall CAD and LAD/LCX/RCA stenosis from routine clinical data and maps those predictions onto an interactive 3D anatomy, with interpretable explanations. CardioLens does this with four design decisions:

- **Use real anatomy, not a cartoon.** The heart wall, every coronary branch, the great vessels and the thorax are BodyParts3D meshes (FMA-identified). Each coronary branch is a separately named mesh bound to a model output, so the vessel the user sees is exactly the vessel the model scored.
- **Make the model agree with anatomy.** In the data, stenosis of any vessel ⇔ CAD (302/303 rows). We therefore model $P(\text{vessel}) = P(\text{CAD}) \cdot P(\text{vessel} \mid \text{CAD})$, which guarantees the 3D view never shows a stenotic artery on a heart the model calls disease-free.
- **Validate the pipeline, not just the classifier.** Algorithm choice, hyper-parameters, calibration, thresholds and conformal quantiles are all learned inside each training fold of repeated nested cross-validation.
- **Explain exactly and admit uncertainty.** Additive models give exact attributions. Every estimate carries a bootstrap interval and a conformal prediction set that says "uncertain" when the model cannot decide.

0.928	**0.849**	**0 / 704**	**1e-9**
CAD ROC-AUC (±0.029)	LAD ROC-AUC (±0.040)	anatomical contradictions, hierarchical vs direct	browser vs server engine max difference

2. Dataset and preprocessing

Data. The UCI *Extension of Z-Alizadeh Sani* dataset (Alizadehsani, Roshanzamir & Sani, 2013; CC BY 4.0) has 303 patients and 59 columns. There are no missing values. Targets: `Cath` (CAD 71.3%) and vessel status for LAD (58.4%), LCX (39.3%) and RCA (37.6%).

Feature registry. `ml/features.py` declares all 54 inputs once, with key, display label, group, type, unit, reference range, plausible bounds and a *modifiable* flag. Groups: demographics 5, history 12, symptoms & exam 13, ECG 7, laboratory 14, echo 3. The training pipeline, the API validator, the input form, the measurements table and the explanation text are all generated from it.

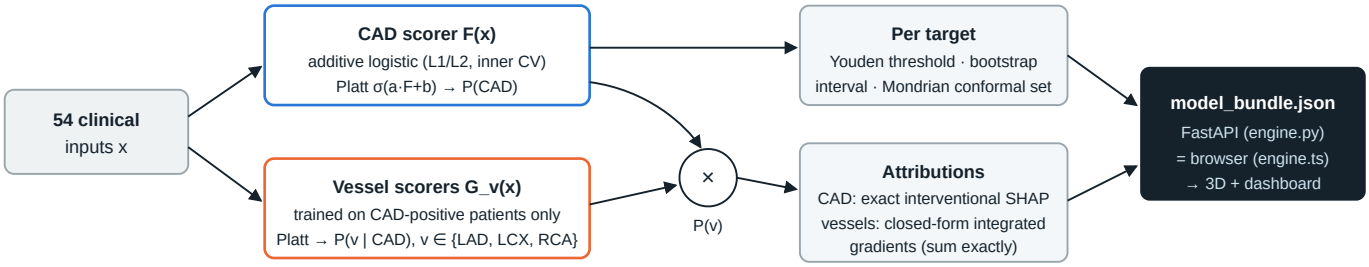
Encoding. Only deterministic row-wise steps happen before the split: Y/N → 1/0, sex → `sex_male`, VHD → ordinal 0–3. `Exertional CP` is constant and dropped. Everything learned from data (standardisation, BBB one-hot) lives inside a scikit-learn `Pipeline`, so it is re-fitted in every fold.

Leakage control. `LAD`, `LCX`, `RCA` and `Cath` are never inputs. This is asserted in the encoder, in the exported bundle and in the test suite. To show why it matters, a model that (wrongly) sees the vessel columns reaches CAD AUC 0.997. A label-permutation test of our pipeline gives AUC 0.492 ± 0.035, so the pipeline does not invent signal.

Label structure. Cath = CAD exactly when ≥1 vessel is stenotic, except for one patient. 86 patients have 0 diseased vessels, 87 have 1, 67 have 2 and 63 have 3. Outer folds are stratified on *CAD* × *number of diseased vessels*, so every fold keeps the multi-label mix.

Reference ranges. Standard adult ranges (e.g. LDL <100 mg/dL, HDL ≥40, FBS 70–100, EF 55–70%) are used only for display and for the "modifiable factors" panel, never as model inputs.

3. Model architecture



Additive scorers. Each target uses a "glass-box" score $F(x) = b_0 + \Sigma_j f_j(x_j)$. With logistic regression each f_j is linear on the standardised input. The family (L1 or L2) and C are chosen by 5-fold inner CV on ROC-AUC. Class weights are balanced, and Platt scaling fitted on inner out-of-fold scores restores calibrated probabilities. The `CardioModel` class also supports Explainable Boosting Machines (piecewise-constant f_j). They were evaluated but not selected (§4).

Hierarchical vessels. P(v | CAD) is trained only on CAD-positive patients, where it is defined, and calibrated there. To fit thresholds and conformal quantiles on all patients without leakage, the product P(CAD)·P(v | CAD) is formed from inner out-of-fold predictions of both factors.

Decision layer. Thresholds maximise Youden's J on inner OOF predictions, which favours sensitivity in a screening setting. **Mondrian (class-conditional) split-conformal** quantiles at α = 0.10 turn each probability into a prediction set: {positive}, {negative} or {both} = *uncertain*. 60 bootstrap refits give a centred 90% interval on every probability.

Explanations. For an additive model, the interventional SHAP value is exact and closed-form: $\varphi_j = a \cdot (f_j(x_j) - E[f_j(X_j)])$. It matches `shap.LinearExplainer` to $4 \cdot 10^{-16}$. For vessels, the log-odds $h = logit(\sigma(L_1)\sigma(L_2))$ is attributed with integrated gradients in term space: $\varphi_j = \Delta_{1j} \int \partial h/\partial L_1 + \Delta_{2j} \int \partial h/\partial L_2$ (48-point quadrature, residual redistributed so contributions sum exactly). Against KernelSHAP on the same reference point, the mean correlation is r = 0.93 and the top-5 features overlap by 89%.

4. Validation methodology and results

Protocol. Repeated stratified 5-fold × 5 outer CV (25 folds) of the *entire* `CardioModel`. Inside each outer training set: 5-fold inner CV for family and C, and separate 5-fold inner OOF predictions for calibration, threshold and conformal quantiles. Metrics are mean ± SD over the 25 outer folds. Curves use pooled out-of-fold predictions. All seeds are fixed (42).

Target	Selected model	ROC-AUC	95% CI*	PR-AUC	Acc.	Prec.	Recall	Spec.	F1	Brier	Coverage
CAD	Logistic L2	**0.928** ± 0.029	0.899–0.956	0.970	0.845	0.947	0.830	0.883	0.883	0.098	90.0%
LAD	P(CAD) × logistic L1	**0.849** ± 0.040	0.792–0.884	0.882	0.771	0.799	0.816	0.712	0.803	0.155	89.4%
LCX	P(CAD) × logistic L1	**0.706** ± 0.054	0.633–0.751	0.578	0.643	0.532	0.810	0.537	0.638	0.207	88.3%
RCA	P(CAD) × logistic L1	**0.738** ± 0.050	0.673–0.792	0.604	0.642	0.517	0.830	0.532	0.633	0.198	89.2%

*Bootstrap CI of pooled out-of-fold AUC (1,000 resamples). Threshold metrics use the in-fold Youden threshold. At a fixed 0.5 threshold, accuracy is CAD 0.865, LAD 0.789, LCX 0.636, RCA 0.658. Coverage is the empirical coverage of the 90% conformal sets (target 90%). Singleton (confident) sets: CAD 90%, LAD 66%, LCX 48%, RCA 50%.

External validation: four other hospitals (920 patients)

Test cohort	n	CAD	ROC-AUC (95% CI)
Cleveland (USA)	304	46%	0.764 (0.71–0.82)
Budapest (Hungary)	293	36%	0.767 (0.71–0.82)
Zurich/Basel (Switzerland)	123	93%	0.702 (0.47–0.91)
VA Long Beach (USA)	200	75%	0.636 (0.55–0.72)
All four, pooled	**920**	**55%**	**0.756 (0.725–0.787)**

Scope: this tests a separate reduced model for overall CAD only, not the deployed 54-input model; LAD/LCX/RCA have no external validation. A reduced CAD model on the 10 inputs both datasets share (age, sex, chest-pain type, resting BP, fasting sugar >120, ECG ST-T change, LVH, total cholesterol ≈ LDL+HDL+TG/5) was trained on all 303 Z-Alizadeh Sani patients and applied unchanged to the UCI Heart Disease cohorts (same label: ≥50% stenosis on angiography). Cleveland's `ca` column is excluded: it is effectively part of the answer. Internal nested-CV AUC of this reduced model: 0.903 ± 0.043. Pooled calibration is near ideal (intercept 0.03, slope 1.00); but the training-site threshold (0.70) is too strict elsewhere: sensitivity 39%, specificity 88% (78% / 63% at 0.5), so a new hospital must set its own threshold. Trained *inside* UCI the same inputs reach 0.839, an upper bound for transfer. Vessels cannot be tested externally (UCI has no per-artery labels).

Why our accuracy is 84.5%, not 95%

Same pipeline, same data	CAD accuracy
Nested, repeated CV (25 outer folds)	**84.5%**
One random 80/20 split, median of 100	85.2%
One split, 5th–95th percentile	78.7–91.8%
One split, luckiest of 100	95.1%
Threshold tuned on the test set (leaky)	96.7%

With 61 test patients, one split's accuracy swings by ±7 points depending on who lands in it; 15% of splits show ≥90%. Reporting a lucky split, or tuning the threshold on the test patients, can produce the 95%+ figures common for this dataset. Every number here comes from folds whose patients were never seen in any modelling choice (`ml/split_optimism.py`).

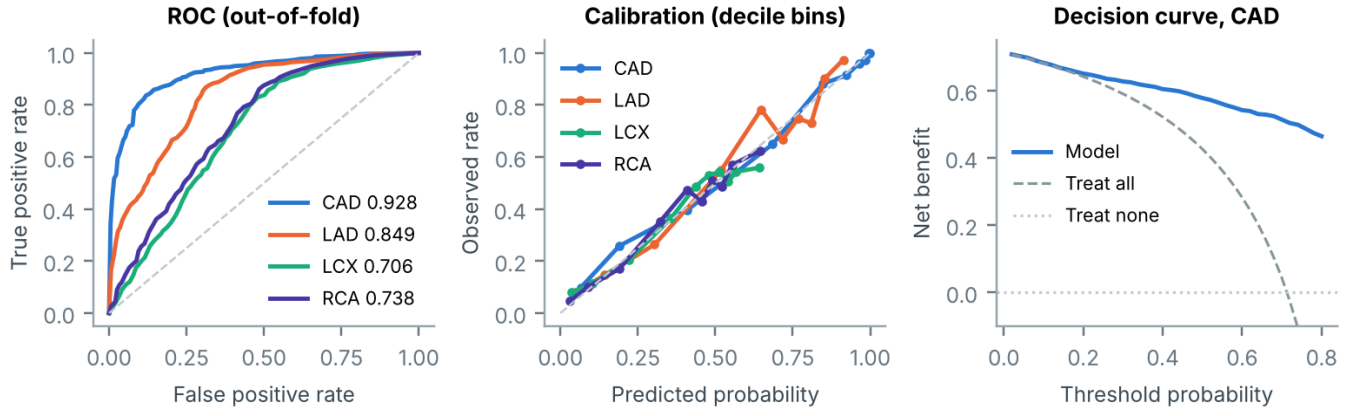


Figure 2. Out-of-fold ROC curves, calibration, and a decision curve for CAD. The model gives positive net benefit over "treat all" across thresholds from about 0.2 to 0.9.

Algorithm comparison (nested CV, ROC-AUC)

Algorithm	CAD	LAD	LCX	RCA
Logistic L2	**0.931**	0.843	0.688	0.730
Logistic L1	0.924	0.838	0.706	**0.742**

Hierarchical vs direct (paired, identical folds)

	ΔAUC (hier − direct)	Brier direct → hier
LAD	+0.009 ± 0.024 (17/25 folds)	0.164 → 0.155

SVM (RBF)	0.922	0.841	0.678	0.722
Random forest	0.925	**0.853**	0.724	0.715
Extra trees	0.924	0.840	0.696	0.732
LightGBM	0.918	0.837	**0.730**	0.723
XGBoost	0.915	0.838	0.728	0.720
CatBoost	0.919	0.841	0.724	0.704
EBM (glass-box GAM)	0.924	0.830	0.676	0.712
Full pipeline, direct vessels	0.928	0.840	0.703	0.746
Full pipeline, hierarchical (deployed)	**0.928**	**0.849**	**0.706**	**0.738**

The first nine rows are single uncalibrated models with direct targets (5×5 nested CV, same seed). Every algorithm is within ±0.01 AUC on CAD/LAD/RCA. LightGBM's LCX 0.730 vs 0.706 is well inside the fold-to-fold SD (±0.054), so it is noise, not a reason to give up exact, browser-portable explanations.

LCX	+0.003 ± 0.050 (13/25)	0.215 → 0.207
RCA	−0.008 ± 0.041 (10/25)	0.199 → 0.198
P(vessel) > P(CAD)	**704 out-of-fold cases (direct) → 0 (hierarchical)**	

Discrimination is equivalent, calibration improves for all three vessels, and anatomical contradictions disappear. That last property matters for a visual system: the colours on the heart are always mutually consistent.

Subgroups (out-of-fold ROC-AUC)

Group	n	CAD	LAD	LCX	RCA
Female / male	127 / 176	0.96 / 0.90	0.85 / 0.83	0.70 / 0.69	0.78 / 0.70
Age < 60 / ≥ 60	162 / 141	0.93 / 0.87	0.88 / 0.72	0.74 / 0.54	0.76 / 0.66
Non-diabetic / diabetic	213 / 90	0.91 / 0.96	0.88 / 0.70	0.71 / 0.61	0.74 / 0.61

Vessel localisation is weaker in older and diabetic patients (diffuse multi-vessel disease), so the UI and README flag it.

5. 3D anatomical pipeline

Source. BodyParts3D 4.0 (© 2008 DBCLS, CC BY-SA 2.1 JP), the resource recommended in the brief, read from a GitHub mirror of its STL files. 85 parts are selected by FMA id in `anatomy/anatomy_registry.json`:

- heart wall (FMA7274)
- left main, LAD and its septal perforators
- LCX
- RCA trunk with its acute marginal, PDA, posterolateral and septal branches
- cardiac veins and coronary sinus
- aorta, venae cavae, pulmonary arteries and veins
- lungs, trachea, diaphragm, sternum, ribs and costal cartilages, clavicles, thoracic spine, body surface

Processing (`build_anatomy.py`):

1. Merge vertices and drop fragments: the heart wall arrives as 2,036 shells.
2. Crop the body surface to the thorax.
3. Quadric-decimate each part to a triangle budget. 3.6 M source triangles become 219 k (heart wall 60 k, coronaries 22 k).
4. Re-orient from BodyParts3D (z-up, +y posterior) to a heart-centred, Y-up, anterior-facing frame. This is a proper rotation, so the patient's left appears on the viewer's right, as in radiology.
5. Export one GLB with named nodes, then compress it with meshopt (4.07 MB → 0.84 MB; the decoder is bundled, so there's no CDN dependency).

Perfusion territories. For every myocardial vertex, distances to point clouds sampled from the LAD, LCX and RCA trees are turned into soft weights $w \propto exp(-(d - d_{min})/6\ mm)$, faded with distance from any epicardial vessel. The weights are stored in the vertex `COLOR_0` attribute. A shader patch on `MeshStandardMaterial` blends the muscle colour toward each territory's current risk colour, in proportion to that risk, so healthy muscle stays muscle-coloured.

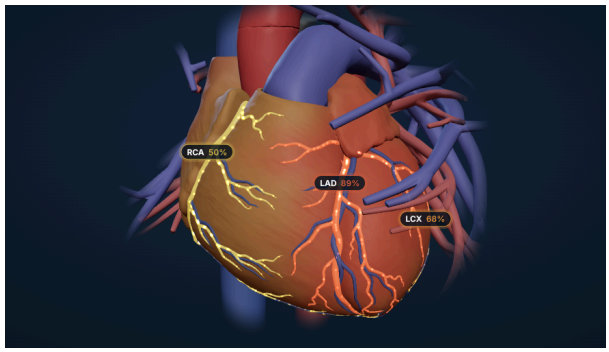
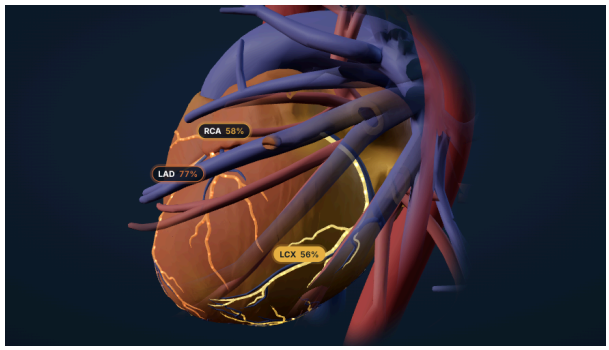


Figure 3a. Heart view. Vessel colour is the per-vessel calibrated probability. Territory tint and bullseye share one colour ramp.



Figure 3b. Thorax view: a Fresnel "ghost" body surface, lungs and skeleton for spatial context.



Centrelines for flow. Each coronary mesh is voxelised (0.6 mm), filled and 3D-skeletonised. The skeleton becomes a graph rooted at the voxel nearest the parent vessel (aorta → LM → LAD/LCX; aorta → RCA → branches). Ostium-to-tip Dijkstra paths with spur pruning give 56 smoothed polylines that carry the flow particles.

Figure 3c. Selecting LCX flies the camera to a left-lateral view and pulses its territory.

Rendering and interaction (React Three Fiber, `HeartScene.tsx`):

- Vessel colour and glow are interpolated each frame toward the latest prediction; uncertain vessels (conformal set = both classes) flicker gently.
- Particles move along the centrelines, slowing as risk rises. This is a visual metaphor, not a haemodynamic simulation, and it is labelled as such.
- The heart beats with a systole/diastole curve at the patient's pulse-rate input.
- Clicking a vessel selects it everywhere (3D, list, bullseye, explanation); hover names the structure. Heart ⇄ Thorax dissolves the thorax with a radial-fade shader.
- Guided tour (camera path with captions written from the prediction), a cross-section clipping plane with shaded chamber walls, an exploded "lift arteries" view, an angiogram-style X-ray look, focus mode for the selected artery, and a synthetic ECG whose morphology follows the ECG inputs.

Performance without a GPU. No shadow maps or HDR environment, a low-power WebGL context, one draw call per part. **Adaptive quality** (drei `PerformanceMonitor`) watches the real frame rate and steps down when it drops: lower render resolution, flow particles off, then a 12 k-triangle myocardium (instead of 60 k) and a paused heartbeat; it steps back up when frames recover. Worst case measured, with *no GPU at all* (SwiftShader software rendering on 2 CPU cores, 1440×900): 2.0 fps at full quality → 4.1 fps in low mode, still usable. `?fps=1` shows a live counter for measuring any machine. If WebGL is unavailable or the 3D context fails, a 2D artery diagram with the same colours replaces the canvas and everything else keeps working.

Honest mapping. The dataset gives vessel-level labels, not lesion positions. CardioLens therefore colours whole vessels and states this on the canvas. RWMA is treated as an input, not a lesion map. The bullseye uses the standard AHA segment → artery assignment and is labelled as the expected ischaemic territory.

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6. System integration and clinical dashboard

Data flow. Every edit re-scores the patient (60 ms debounce): `POST /api/predict` computes in under 1 ms, or the in-browser engine in about 1–9 ms. The result drives the 3D materials and every dashboard panel at once.

One model, two engines. `train_final.py` exports the fitted pipeline as `model_bundle.json`: additive terms, Platt parameters, thresholds, conformal quantiles and 60 bootstrap models. `ml/engine.py` (server) and `engine.ts` (browser) implement the same arithmetic. Tests check that sklearn = Python engine = TypeScript engine on all 303 patients for probabilities, intervals, sets and attributions, to 1e-9.

API (FastAPI, OpenAPI docs at `/docs`):

- `/api/predict` validates every input against the feature registry. It rejects unknown keys and invalid categories with 422, warns on out-of-range values, and fills omitted inputs with the median patient and lists them.
- `?check=true` re-scores through the original sklearn pipelines; `/api/whatif` returns before/after/delta; `/api/predict/batch` and `/api/meta` complete the interface.

Extensibility. Three registries decouple the layers: `ml/features.py` (clinical inputs), `config/registry.json` (targets, their meshes and AHA segments) and `anatomy/anatomy_registry.json` (meshes by FMA id, with an optional `target` binding). A new lab value, vessel target or anatomical structure is a registry entry plus a re-run; the UI and API need no code changes.

Two views. A *guided view* walks beginners through five steps (Patient → Heart → Why → What if → Report), with a narrated 3D tour, glossary tooltips on every technical term, and a one-click PDF report. An *expert view* shows every panel at once.

Dashboard components (brief §3):

- **Assessment:** CAD probability with band, 90% interval and conformal verdict. Per-vessel bars show the probability fill, interval whisker and operating threshold.
- **AHA 17-segment bullseye** coloured by vessel risk and linked to the 3D selection.
- **Why this estimate:** a generated plain-language sentence, a SHAP force plot, odds-multiplier reason cards, a body-system breakdown and a feature × target contribution heatmap (the dual-encoded format clinicians preferred in a 206-clinician study), for CAD or any vessel.
- **Measurements:** every physiological value next to its reference range (with high/low flag), its signed contribution and its share of the total attribution (brief §3c).
- **Modifiable factors:** re-scores with one out-of-range factor moved to its range limit, with one-click "apply". Labelled as model sensitivity, not treatment effect. Changes that move the wrong way in this data (e.g. lowering LDL or BMI, plausibly reflecting treatment of known disease) are never offered as advice; they are collapsed under a labelled "dataset artifact" note.
- **Clinical-note import** (a second input modality): paste or open a referral letter or lab report; transparent rules extract up to 54 inputs in the browser (negations such as "no Q

waves", SI units such as mmol/L converted) and show the source words for each value before loading.
- **Similar patients:** the 12 nearest dataset patients (importance-weighted distance) with their actual angiography results and out-of-fold scores.
- **Model evidence** (all of §4, inside the app) and **Download report**: a 2-page A4 PDF built in the browser (3D snapshot, estimates, reasons, measurements, validation, disclaimer), plus CSV and JSON exports.

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7. Usage

```
make install      # pinned Python + Node dependencies
make reproduce    # retrain (nested CV), external validation, split study, then check all 20 reported metrics
(±0.005) + tests
make test         # 23 tests: no-leakage (LAD/LCX/RCA/Cath never inputs), engine parity to 1e-9, API contract
make e2e          # 19 browser checks: expert + guided journeys, note import, both PDFs, no-WebGL fallback,
server = browser
make serve        # FastAPI + app → http://localhost:8000      (or: docker compose up --build)
make single       # CardioLens3D.html: the whole app, model and 3D heart in one offline file (hosts anywhere)
```

Workflow: pick a dataset patient (its angiography result is shown), a preset, or edit any value → rotate the heart and click a vessel → read the verdict and explanation → try modifiable factors → check similar patients → export the report.

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8. Limitations and responsible use

- Training data are 303 patients from one centre. External validation of overall CAD on 920 patients from four other hospitals shows useful but lower discrimination (AUC 0.76) and a threshold that needs local recalibration; per-vessel predictions are validated internally only. Probabilities reflect a 71% CAD prevalence, far above a screening population.
- LCX and RCA are only moderately predictable from these features (AUC 0.71–0.74). Conformal sets make this visible: about half of vessel predictions are flagged "uncertain".
- Territories come from a single reference heart (coronary dominance varies) and flow particles are illustrative. Associations are not causal; the modifiable-factors panel is framed as model sensitivity.

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9. Credits

Dataset: Alizadehsani R., Roshanzamir M., Sani Z. (2013). *Extention of Z-Alizadeh Sani dataset.* UCI Machine Learning Repository, doi:10.24432/C5461K, CC BY 4.0.
Anatomy: BodyParts3D, © 2008 Life Science Integrated Database Center, CC BY-SA 2.1 Japan; derived meshes are shared under the same licence.
Standards: Cerqueira M.D. et al., AHA 17-segment model, *Circulation* 2002;105:539–542. **Software:** scikit-learn, interpret, SHAP, LightGBM/XGBoost/CatBoost, FastAPI, three.js / React Three Fiber, trimesh, meshoptimizer. **Code:** MIT licence.